SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 1

Answer ALL questions.

1. What are the four basic operations that can be performed on an array, and how does the number of steps vary for each operation?

2. Explain why reading from an array is considered an efficient operation. How does the computer locate an element at a specific index in just one step?

3. Compare the efficiency of searching for an element in an array using linear search. What is the maximum number of steps required for a linear search on an array with N elements?

4. What is the difference between inserting an element at the end of an array versus inserting an element in the middle of an array? How does this affect the time complexity?

5. How does the time complexity of deletion from an array differ when deleting an element at the beginning, middle, or end of the array? Describe the steps involved in the worst-case scenario.